MuGyum **Kim**

AI/ML Undergraduate Researcher · Computer Vision & Generative Models

Daejeon, Republic of Korea

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Education

Chungnam National University — *Daejeon, Republic of Korea*

B.Eng. in Artificial Intelligence — *Mar. 2022 – Feb. 2028*

- Major GPA 4.39/4.5 | Cumulative GPA 4.22/4.5

University of Illinois Urbana-Champaign, Grainger College of Engineering — *Urbana-Champaign, IL, USA*

Undergraduate Summer School · AI and Generative Models (Certificate of Recognition) — *Jul. 2026 – Aug. 2026*

- Reimplemented DDPM on MNIST from scratch; class conditioning raised label agreement with the source digit from 0.700 to 0.993 while barely reducing pixel distance, and RePaint cut outside-mask error from 0.701 to 0.005.

Publications

MuGyum Kim*, GiWon Kim, ChanJong Park, SungHo Lee†. "Aggregation Sensitivity in Diffusion Inpainting Along a Mean–Median Axis." *IEEE/IEIE International Conference on Consumer Electronics–Asia (ICCE-Asia) 2026*, pp. 396–401. **Oral.**

*Presenting author. †Corresponding author.

Research Experience

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · Funded by Saemmulter, Inc. — *Sep. 2026 – Oct. 2026*

- Re-evaluated prior-work TCN and DLSTM groundwater-level forecasters against a persistence baseline on 58 wells (416 days of hourly data), fixing every configuration on a development period before opening an 84-day test period once.
- Found no TCN–DLSTM difference that survived Holm correction in any of six conditions; recommended TCN by a pre-specified parameter-count rule, with per-well checks against persistence before deployment.
- Delivered the final report, a single-file HTML dashboard, and a reproducibility package (18 training and 184 inference reruns bit-identical).

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · Funded by ETRI — *Mar. 2026 – Sep. 2026*

- Built a firewall flow-log detector that groups each host's flows into packet-count-triggered epochs (Scan Detection Sketch, $\theta = 20$), so features do not depend on traffic rate, and extracts 28 behavioral features per epoch.
- Trained a multi-label MLP for normal, scanning, C2 beacon, and malware-delivery traffic, reaching 94.8% test macro-F1 on source-IP-disjoint splits (95.3% validation).
- Ran inference on 16.2M unlabeled production epochs, diagnosed the prediction distribution, and deployed scanning detection as a ClickHouse-backed batch and real-time service.

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher — *May 2023 – Feb. 2024*

- Curated and labeled an 81,000+ sample satellite imagery dataset from an industry partner for cropland and greenhouse segmentation.
- Implemented and trained a SegFormer-based segmentation model, achieving 88.9% mIoU on a three-class cropland classification task.
- Built and labeled a vehicle pose estimation dataset from real-world CCTV footage across multiple environments, supporting a drunk-driving analysis pipeline.

Skills

Languages & ML Python, C/C++, PyTorch, segmentation-models-pytorch, timm

Libraries & Tools NumPy, Pandas, scikit-learn, OpenCV, ClickHouse, FastAPI, Weights & Biases, Git

Honors & Activities

2022–23 **Academic Excellence Scholarship**, Chungnam National University (2022/2, 2023/1, 2023/2)

2023 **SW-Centered University Activity Scholarship**, Chungnam National University

2026 **Undergraduate Tutor, "AI and Future Society"**, attendance, student Q&A, and midterm proctoring support

2023 **Treasurer, DIVE (AI Club)**, organized a corn seed classification competition